

FASSBERG, Germany

WMO: 102460

Lat: 52.9156N

Lon: 10.1886E

Elev: 288

StdP: 14.54

Time Zone: 1.00 (EUC)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.3	16.2	7.5	8.2	13.2	12.6	10.6	17.8	25.3	44.9	22.8	43.4	6.1	100	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	86.2	66.7	82.4	65.2	78.7	63.6	68.4	81.3	66.8	78.9	65.2	75.9	8.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.4	91.6	71.8	62.7	86.1	70.7	61.0	81.0	68.9	32.8	81.1	31.5	79.2	30.2	75.3	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	18.1	16.0	4.9	93.4	7.6	4.0	-0.6	96.2	-5.1	98.5	-9.4	100.8	-14.9	103.7
			4.5	72.2	7.5	2.2	-0.9	73.8	-5.3	75.1	-9.5	76.4	-14.9	78.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	32.9	35.0	39.3	47.2	54.6	60.2	64.2	63.7	56.5	48.9	40.6	35.1	(6)
	DBStd	13.09	8.86	8.18	7.08	7.08	6.85	5.99	5.76	5.53	5.70	6.65	6.89	8.46	(7)
	HDD50	2314	530	422	337	136	29	1	0	0	9	101	287	462	(8)
	HDD65	6262	995	840	797	535	330	169	86	91	263	500	731	925	(9)
	CDD50	1680	0	2	5	51	172	306	441	426	203	67	6	1	(10)
	CDD65	153	0	0	0	0	8	25	62	52	6	0	0	0	(11)
	CDH74	2026	0	0	0	13	156	375	742	628	108	4	0	0	(12)
	CDH80	600	0	0	0	1	31	107	238	200	23	0	0	0	(13)
(14) Wind	WSAvg	7.4	8.1	8.3	8.3	7.5	7.4	7.1	7.0	6.4	6.5	7.2	7.2	7.9	(14)
(15) Precipitation	PrecAvg	30.6	3.0	2.2	2.2	1.8	2.3	2.7	3.0	2.9	2.4	2.4	2.6	3.0	(15)
	PrecMax	41.0	6.2	4.8	5.2	3.8	6.4	5.2	7.2	5.9	7.8	7.3	5.0	6.7	(16)
	PrecMin	21.8	0.2	0.6	0.3	0.4	0.7	1.1	0.8	0.9	0.6	0.8	0.1	1.0	(17)
	PrecStd	5.1	1.6	1.1	1.3	0.8	1.3	1.1	1.5	1.3	1.6	1.3	1.3	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.3	57.5	65.5	75.4	84.0	89.5	91.4	91.2	84.0	71.6	58.9	55.1	(19)
		MCWB	49.3	46.7	51.9	58.2	64.2	69.2	67.2	67.9	64.8	60.9	54.1	51.0	(20)
	2%	DB	49.9	52.9	59.2	69.9	78.5	82.7	86.2	85.5	76.9	65.8	55.5	51.8	(21)
		MCWB	46.6	46.9	49.0	55.2	61.9	65.5	67.1	66.1	63.0	58.4	52.0	49.1	(22)
	5%	DB	46.9	49.6	54.1	66.2	73.7	78.2	82.0	80.9	71.7	62.0	52.9	48.6	(23)
		MCWB	44.5	45.6	47.0	52.9	58.8	63.0	65.4	65.1	60.4	56.7	49.4	46.4	(24)
	10%	DB	44.4	46.7	50.5	61.9	69.6	73.7	77.5	76.6	67.8	58.9	50.2	46.3	(25)
		MCWB	42.2	43.1	45.4	51.7	57.3	60.8	63.5	63.5	59.0	55.0	47.6	44.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.1	50.1	53.4	60.0	66.6	70.4	71.0	70.5	66.7	62.1	56.0	51.9	(27)
		MCDB	51.8	54.0	62.0	71.5	80.2	83.5	83.4	83.6	78.8	68.6	58.3	54.5	(28)
	2%	WB	47.2	47.8	50.7	56.9	63.3	67.3	68.3	68.2	63.9	59.4	52.5	49.4	(29)
		MCDB	49.3	51.7	56.3	66.4	75.4	79.8	81.5	80.1	73.3	64.2	55.2	51.1	(30)
	5%	WB	44.7	45.4	48.4	54.7	60.6	64.6	66.7	66.5	61.9	57.3	50.0	46.8	(31)
		MCDB	46.7	49.0	53.4	63.9	70.9	74.8	79.1	77.6	69.8	61.3	52.0	48.7	(32)
	10%	WB	42.1	42.9	45.9	52.4	58.1	62.4	65.0	64.8	60.0	55.2	47.7	44.0	(33)
		MCDB	43.9	46.4	50.1	60.5	67.1	71.8	75.6	74.3	66.2	58.7	49.8	45.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	11.9	15.5	21.2	21.1	20.9	19.8	20.7	19.2	14.7	10.1	8.7	(35)
		MCDBR	10.9	15.0	23.6	28.8	28.8	28.8	27.9	28.4	26.4	19.1	11.8	9.7	(36)
	5% WB	MCWBR	9.1	10.3	15.0	15.9	14.9	14.2	12.1	12.6	14.5	12.8	9.0	8.6	(37)
		MCWBR	9.8	12.6	19.9	25.0	25.4	25.3	24.3	24.4	22.1	17.7	10.7	9.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.315	0.335	0.367	0.410	0.408	0.406	0.423	0.416	0.385	0.353	0.332	0.309	(40)
		taud	2.369	2.363	2.295	2.218	2.265	2.292	2.254	2.289	2.364	2.437	2.411	2.360	(41)
	Ebn at Noon		212	244	258	260	267	269	261	256	250	234	203	191	(42)
		Edh at Noon	19	25	33	40	40	39	40	37	30	23	18	16	(43)
(44) All-Sky Solar Radiation	RadAvg		195	415	768	1265	1568	1670	1578	1323	969	544	239	149	(44)
	RadStd		21	57	77	157	182	130	218	103	112	59	25	11	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	-275	N/A	N/A

Nomenclature: See separate page